Green Skills

The big idea

Everything around us – air, water, soil, plants, animals – forms our **environment**. Humans depend on it for food, water and air, yet our activities often damage it. **Green skills** are the knowledge and habits that help us protect the environment while living and working.

Environment and ecosystem

- **Environment**: all living and non-living things around us and their interactions.
- **Ecosystem**: a community of living organisms interacting with each other and with their non-living surroundings in a particular area, e.g. a pond, a forest, a desert.

COMPONENTS OF AN ECOSYSTEM

- **Biotic (living)**: producers (green plants), consumers (herbivores, carnivores, omnivores) and decomposers (bacteria, fungi).
- **Abiotic (non-living)**: sunlight, air, water, soil, temperature, minerals.

Food chain and energy flow

A **food chain** shows who eats whom: **Grass → Grasshopper → Frog → Snake → Eagle**. Energy flows from the Sun to producers and then to consumers. **Decomposers** break down dead matter and return nutrients to the soil, so materials are recycled. Many inter-connected food chains form a **food web**.

Natural resources

Type	Meaning	Examples
Renewable	Replenished naturally	Sunlight, wind, water, forests (if used wisely)
Non-renewable	Limited; take millions of years to form	Coal, petroleum, natural gas, minerals

Threats to the environment

- **Pollution** – air (vehicle and factory smoke), water (sewage, chemicals), soil (pesticides, plastic), noise.
- **Deforestation** – loss of forests and wildlife habitat.
- **Overuse of resources** – groundwater depletion, overfishing.
- **Global warming and climate change** – caused by greenhouse gases like carbon dioxide and methane.

OZONE HOLE ≠ GLOBAL WARMING

Ozone depletion (caused mainly by CFCs) lets more UV rays reach Earth. Global warming is heating of Earth due to greenhouse gases. They are different problems.

Conservation of natural resources

- **3Rs**: **Reduce** use, **Reuse** items, **Recycle** waste.
- Save electricity and water; use solar energy.
- Plant trees; protect forests and wildlife.
- Rainwater harvesting.
- Use public transport, cycle or walk.
- Segregate wet and dry waste; compost kitchen waste.

Green economy and green jobs

A **green economy** aims at growth with low carbon emissions, efficient use of resources and social inclusion. **Green jobs** protect or restore the environment, in sectors like renewable energy, organic farming, waste management, water conservation, green construction and eco-tourism.

GREEN JOBS AROUND YOU

Solar panel installer, organic farmer, waste recycler, rainwater harvesting technician, forest guard.

EXAM TIP

A typical 2–mark question: "Differentiate between biotic and abiotic components with two examples each." Learn clear examples.

Quick recap

- Environment = all surroundings; ecosystem = living + non-living interacting in an area.
- Biotic: producers, consumers, decomposers. Abiotic: sun, air, water, soil.
- Renewable vs non-renewable resources.
- Threats: pollution, deforestation, global warming. Remedies: 3Rs, saving energy and water, planting trees.
- Green economy and green jobs protect the environment.

Practice questions

1. Which of these is an abiotic component of an ecosystem?
   (a) Grass (b) Fungi (c) Sunlight (d) Frog
2. Bacteria and fungi that break down dead matter are called
   (a) producers (b) herbivores (c) carnivores (d) decomposers
3. Which is a non-renewable resource?
   (a) Coal (b) Wind (c) Sunlight (d) Water
4. In the food chain Grass → Grasshopper → Frog → Snake, the producer is
   (a) grasshopper (b) grass (c) frog (d) snake
5. Which practice helps conserve resources?
   (a) Leaving lights on all day (b) Burning plastic waste (c) Rainwater harvesting (d) Cutting trees for fuel
6. Which of these is a green job?
   (a) Oil well driller (b) Coal power plant operator (c) Plastic packaging seller (d) Organic farmer
7. Differentiate between renewable and non-renewable resources with two examples each.
8. What is an ecosystem? Explain its biotic and abiotic components with examples.

Answers: 1. c 2. d 3. a 4. b 5. c 6. d 7. see notes 8. see notes